Bayesian Optimization of Nonlinear Half-Car Models for Passenger Comfort over Road Disturbances

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1. Introduction

Designing comfortable and safe suspension systems for vehicles across diverse road conditions is a complex engineering challenge faced by modern automotive manufacturers. A car's suspension must strike a balance between ride comfort and dynamic performance, especially when intended for commercial use where passenger experience is a priority.

Figure 1. Toyota Prius – the most sold car in 2017 (Press, 2015).

In this study, we focus on the Toyota Prius (Figure 1) as a case study vehicle. Using constructor-estimated values, we evaluate how this car responds to a variety of road disturbances. The goal is to assess and enhance passenger comfort by modeling and optimizing its suspension system.

Engineering Context We simulate a classic physical-world problem: the half-car suspension model, in both its linear and non-linear configurations. This abstraction captures vertical and pitch dynamics of a vehicle, making it suitable for studying the impact of road profiles such as sinusoidal inputs and realistic speed bumps. By recreating road scenarios that vehicles routinely encounter, this model allows for in-depth exploration of transient vehicle behavior and ride comfort.

Problem Statement A central challenge in vehicle engineering is balancing competing objectives in suspension design. These include maximizing ride comfort, maintaining sufficient suspension travel, controlling body attitude (pitch and roll), and managing dynamic tire loads. These performance goals often conflict, and the appropriate trade-offs depend heavily on the intended application of the vehicle—whether for domestic use, commercial transport, or high-performance racing (Figure 2).

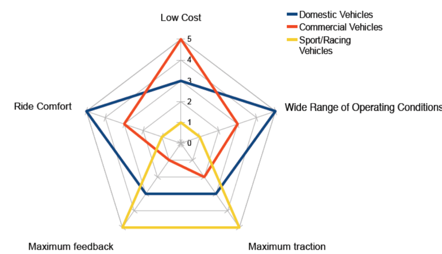


Figure 2. Radar plot illustrating trade-offs between key design characteristics for domestic, commercial, and sport/racing vehicles (Darling, 2023).

Traditional optimization techniques in suspension design typically overlook model uncertainty and require large numbers of simulations or experiments to reach acceptable solutions. This makes them less practical for real-world use, where road inputs and passenger comfort expectations vary significantly. In this study, we explore how machine learning—specifically, Bayesian Optimization (BO) — can offer a more efficient and uncertainty-aware method for tuning suspension parameters to enhance passenger comfort across realistic driving conditions.

Contributions This paper makes three key contributions:

1. **Complete Simulink Implementation:** A full Simulink model of a simplified half-car system is developed from the ground up, incorporating both linear and non-linear suspension dynamics. [1] No pre-existing templates or libraries were used in its construction.

2. **Test Bench Framework:** A critical contribution is our use of systematic test benches to verify the validity of each subsystem (body, suspension, wheel) before any optimization is performed. These tests are essential to ensure the simulated model reflects theoretical expectations and behaves as a physically plausible system.

3. **Comfort-Oriented Optimization:** Using road profiles

[1] The model intentionally excludes components such as anti-roll bars and complex tyre behavior (e.g., Magic Formula) to maintain simplicity and focus on vertical and pitch dynamics.

including sinusoidal disturbances and UK-standard speed bumps (e.g., Watts and Seminole humps), we simulate the suspension response. We then quantify ride comfort using ISO 2631-4 standards via RMS acceleration metrics and before optmizing, we follow a sensitivity analysis to identify which parameters most influence comfort. This is followed by a BO to tune car's parameters (damper and spring parameters) for maximal passenger comfort.

Motivation and Broader Impact This study serves as a proof-of-concept for applying modern machine learning techniques to traditional mechanical engineering problems. By addressing a real-world application—enhancing passenger experience in commercial vehicles—we demonstrate the potential of data-driven design and optimization in physical-world systems.

2. Background

2.1. Half-Car Modeling

This section aims to layout the mathematics of kinematics and vehicle dynamics equations for the half car model. The full MATLAB Simulink implementation can be seen in Figure 26 in Appendix C and the reader is encouraged to refer to Table 3 for the list of symbols used throughout this study.

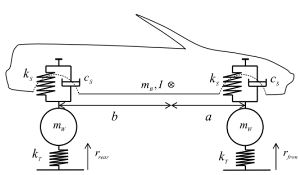


Figure 3. Simplified half-car model showing two wheels connected by the car body with suspension elements (k_s, c_s) and tire stiffness (k_t). The body mass m_b rotates about its center (θ), with distances a and b to the front and rear axles. This model is used to study the vehicle's dynamic response to road disturbances.

Half-Car Body Subsystem The half-car model (Figure 3) can be divided into various subsystems, with the body subsystem being one of them. In our analysis, we will employ Newton's second law to scrutinize forces and kinematics, allowing us to derive equations to both linear aspects (resolving forces: Eq 1) and rotational aspects (resolving moments: Eq 2).

$$M_b \ddot{S}_b = F_f + F_r - M_b g \tag{1}$$

$$I\ddot{\theta} = aF_f + bF_r \tag{2}$$

From this point, we can deduce equations for the velocity and displacement of both the front and rear portions of the body subsystem:

$$\dot{S}_{b,f} = \dot{S}_b + a\theta \tag{3}$$

$$\dot{S}_{b,r} = \dot{S}_b - b\theta \tag{4}$$

$$S_{b,f} = S_b + a\theta \tag{5}$$

$$S_{b,r} = S_b - b\theta \tag{6}$$

Half Car Wheels Subsystems We can further deduce the wheel models by resolving the equation of motion vertically (Eq 2):

$$M_W \ddot{S}_W = -F + K_T(r - S_W) - M_W g \tag{7}$$

The physical implementation of the wheel module (Fig 29) has been realized in MATLAB Simulink:

Half Car Suspension Subsystems Additionally, when we resolve the suspension force vertically, we acquire the following:

$$F = -K_s(S_B \frac{f}{r} - S_W \frac{f}{r}) - C_s(\dot{S}_B \frac{f}{r} - \dot{S}_W \frac{f}{r}) \tag{8}$$

The suspension module (Fig 31) has been physically implemented in MATLAB Simulink:

2.2. Non-Linear Properties

To enhance the realism of our half-car model and better simulate real-world vehicle dynamics, we extend the suspension system to include non-linear characteristics. While linear models provide foundational insights, they fall short in accurately capturing effects such as spring hardening and asymmetric damping, which are prominent in physical suspension systems.

Spring Hardening Effect The linear spring force is typically defined as:

$$f = kx, \tag{9}$$

where k is the linear stiffness coefficient and x is the displacement from the equilibrium position.

To model non-linear stiffness, we introduce a spring hardening term that activates once the displacement exceeds a threshold x_0. The resulting piecewise force-displacement relationship is given by:

$$f(x) = \begin{cases} kx, & \text{for } x \leq x_0 \\ kx + k_2(x - x_0), & \text{for } x > x_0 \end{cases} \tag{10}$$

This formulation adds an additional stiffness k_2 beyond the threshold x_0, effectively increasing the restoring force at larger displacements. As illustrated in Fig. 4, the transition occurs smoothly at x_0, and the total force beyond that point grows more steeply. This captures the stiffening behavior of materials or suspensions under high loads.

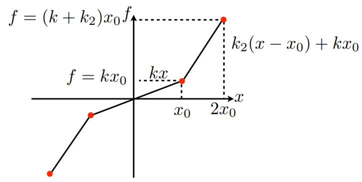


Figure 4. Spring hardening behavior modelled using a piecewise linear function.

Asymmetric Damping In addition to spring non-linearity, we model damping asymmetry between extension and compression phases. The damping force f_d is defined as:

$$f_d = \begin{cases} -C_{\text{ext}} \cdot \dot{x}, & \text{if } \dot{x} > 0 \text{ (extension)} \\ -C_{\text{comp}} \cdot \dot{x}, & \text{if } \dot{x} \leq 0 \text{ (compression)} \end{cases} \tag{11}$$

where C_{ext} and C_{comp} are the damping coefficients for extension and compression, respectively.

2.3. Suspension Design and Passenger Comfort

The vehicle suspension system plays a critical role in isolating passengers from road-induced vibrations, maintaining continuous tyre-road contact, and controlling pitch and roll. These dynamics are governed by two main components:

Springs store mechanical energy from road disturbances, limiting vibration transfer to the chassis. Energy must be stored within safe stress levels to preserve material life. Coil and torsion springs typically offer better weight-to-performance ratios than leaf springs.

Dampers regulate how quickly this stored energy is dissipated. Suspension tuning involves trade-offs: softer damping improves ride comfort but may compromise handling; stiffer damping enhances road grip but can reduce passenger comfort.

The primary goal of suspension design is to minimize vertical accelerations transmitted to the body—particularly in the 4–8 Hz range—where human sensitivity is highest. This band corresponds to the natural frequencies of internal organs and the spinal column (see Table 1).

Comfort guidelines from ISO 2631-4:2001 (ISO, 2021) suggest that bounce frequencies between 1.5–2.3 Hz best match human gait and promote comfort. In contrast, vibrations above 20 Hz are usually well-damped, though lateral and pitch disturbances can still be discomforting if aligned with body resonances.

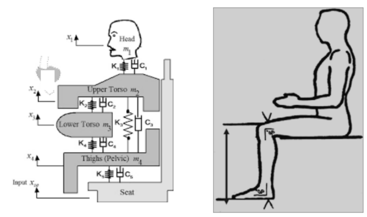


Figure 5. Biomechanical free-body diagram of human vibration sensitivity across body regions.

Table 1. Typical Natural Frequencies of Human Body Components (Ren et al., 2018)

Body Part	Natural Frequency
Skull	300–400 Hz
Lower Jaw-Skull	100–200 Hz
Eyeball	50–90 Hz
Head-Neck-Shoulder	20–30 Hz
Spinal Column	10–12 Hz
Thorax-Abdomen	4–8 Hz

2.4. Road Profile Modeling and Excitation Analysis

To design a suspension system aligned with human comfort metrics, road excitation must be realistically modeled. Road surfaces are represented using a stochastic model defined by their Power Spectral Density (PSD), $S(n)$, in units of $\mathrm{m}^2/(\mathrm{cycle/m})$. The PSD describes the amplitude distribution of surface roughness over spatial frequency n, and typically follows:

$$\log S(n) = \log C - m \log n, \quad \text{with } m = 2.5 \tag{12}$$

Here, C depends on surface type ($C = 10^{-7}$ for motorways, higher for rougher roads). The RMS amplitude of the road disturbance is estimated by:

$$\text{Amp} = \left(\int_{n_1}^{n_2} S(n)\, \mathrm{d}n \right)^{1/2} \tag{13}$$

Given a vehicle speed v, spatial frequency n is converted into temporal excitation frequency via $f = v \cdot n$. This frequency directly influences how the car body and passengers respond to road inputs.

Design Implications: Reducing excitation within the 4–8 Hz band is critical to minimizing discomfort. Suspension design must thus be optimized to attenuate input frequencies that overlap with human resonance bands. Additionally, larger wheels may reduce sensitivity to sharp edges but increase unsprung mass, altering handling dynamics. Effective modeling of road input and human perception is therefore essential to achieving a high-fidelity comfort assessment.

2.5. Sensitivity Analysis

Monte Carlo Sampling Monte Carlo sampling employs random sampling of input parameters to evaluate the model's output distribution. By performing numerous simulations with inputs drawn from specified probability distributions, it estimates the impact of input uncertainties on the output. This method is particularly useful for capturing the effects of parameter uncertainties on system behavior.

Sobol Sampling Sobol sampling is a variance-based global sensitivity analysis technique that decomposes the output variance into contributions from each input parameter and their interactions. It quantifies the influence of individual inputs and their combinations on the output variance through Sobol indices, providing a comprehensive understanding of parameter importance in non-linear and non-additive systems.

2.6. Bayesian Optimization

To efficiently optimize the suspension parameters for ride comfort, we utilize BO, a strategy well-suited for expensive-to-evaluate functions. We have used the `bayespot` Matlab library.

Objective Function The primary goal is to minimize the RMS value of the vertical acceleration experienced by the vehicle's body, which serves as a quantitative measure of ride comfort. Mathematically, the objective function $J(\mathbf{x})$ can be defined as:

$$J(\mathbf{x}) = \sqrt{\frac{1}{T}\int_0^T a_z(t;\mathbf{x})^2\,dt} \tag{14}$$

where:

- $\mathbf{x}$ represents the vector of suspension parameters to be optimized (e.g., spring stiffnesses, damping coefficients).
- $a_z(t;\mathbf{x})$ denotes the vertical acceleration of the vehicle body at time t, which is a function of the parameters $\mathbf{x}$.
- T is the total duration of the simulation or measurement.

This formulation aligns with standard practices in suspension optimization, where the RMS of vertical acceleration is minimized to enhance passenger comfort (Savaia et al., 2021).

Surrogate Model BO relies on a surrogate model to approximate the objective function. We use Gaussian Process Regression (GPR) due to its ability to provide both mean predictions and uncertainty estimates. The GPR model is defined by:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \tag{15}$$

where $m(\mathbf{x})$ is the mean function and $k(\mathbf{x}, \mathbf{x}')$ is the covariance function. This probabilistic model allows for the estimation of the function's behavior across the parameter space (Frazier, 2018).

Acquisition Function The acquisition function guides the selection of the next sampling point by balancing exploration and exploitation. Common acquisition functions include:

- **Expected Improvement (EI)**: Focuses on areas with high potential for improvement over the current best observation.
- **Probability of Improvement (PI)**: Targets regions with a high probability of outperforming the current best.
- **Upper Confidence Bound (UCB)**: Balances the mean prediction and uncertainty to explore uncertain regions.

In our study, we employ the Expected Improvement criterion, defined as:

$$\mathrm{EI}(\mathbf{x}) = \mathbb{E}[\max(f_{\text{best}} - f(\mathbf{x}), 0)] \tag{16}$$

where f_{best} is the best objective value observed so far.

3. Methodology

3.1. Test benches

Each subsystem was individually tested using standard benchmark scenarios to validate the physical accuracy of the model. As test benches are a fundamental component of simulation design, detailed results are provided in Appendix B for reference.

Body Subsystem Two core tests were performed:

- **Fall Under Gravity**: With no external forces, the body exhibits a parabolic displacement trajectory, aligning with $S_B = -\frac{1}{2}gt^2$.
- **Balance Weight Test**: Opposing forces equal to body weight are applied ($F_f = F_r = -M_b g$), yielding zero net displacement and velocity.

Wheel Subsystem Three tests verified wheel behavior:

- **Fall Under Gravity** and **Balance Weight Test**: Similar to body results due to shared vertical dynamics.
- **Step Input Test**: Produces oscillations at $f = \frac{1}{2\pi}\sqrt{K_t/M_w} \approx 13.315$ Hz, matching analytical calculations.

Suspension Subsystem An I/O force test was conducted on 16 binary combinations of displacement and velocity inputs, confirming exact force outputs for a linear suspension with $K_s = 20{,}000$ N/m, $C_s = 1000$ Ns/m.

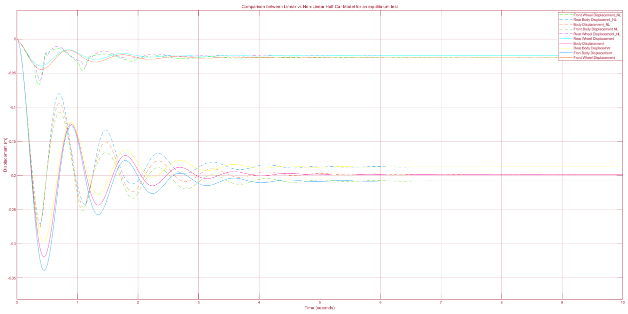
Half-Car System Tests System-level responses were evaluated:

- **Balance Test**: Body and wheel displacements at equilibrium match analytical results based on load distribution and stiffness.
- **Wheel Bounce Test**: Large stiffness and damping simulate rigid suspension behavior, confirming dynamic consistency.
- **Spring Bounce Test**: High K_t and near-zero M_w produce expected oscillatory body response.

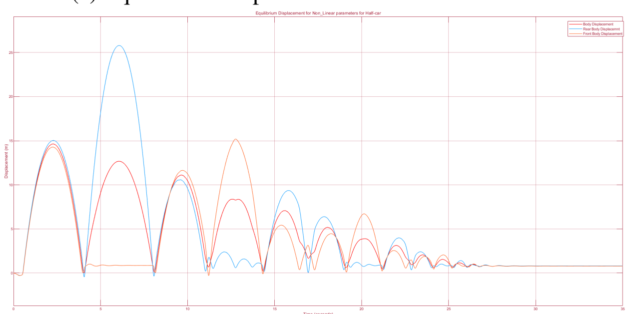
Non-Linear Suspension Characteristics To better capture real-world behavior, we extended the model to incorporate non-linear damping and stiffness characteristics. These were implemented in Simulink using Lookup Tables for damping asymmetry (extension vs. compression) and spring hardening effects.

After verifying correct behavior of these lookup components (see Appendix B), the non-linear configuration was subjected to the same test benches used for the linear model. The most notable differences appeared in transient response:

- In the **Balancing Test** (Figure 6a), both linear and non-linear models converged to the same steady-state, but the non-linear model exhibited reduced peak oscillations due to variable damping and stiffness.
- In the **Step Input Test** (Figure 6b), the non-linear model showed a wheel detachment event, revealing the heightened sensitivity of the system to sudden excitation when non-linear effects are present.



(a) Equilibrium displacement: linear vs. non-linear



(b) Wheel detachment under step input

Figure 6. Non-linear model behavior: steady-state comparison and step input response.

3.2. Vehicle's transient response

To evaluate suspension performance under realistic driving conditions, we simulated the vehicle's response to a sinusoidal road profile with a 1m wavelength and varying amplitudes. The road input is defined as:

$$r = h \sin\left(2\pi \frac{v}{3.6} t + 2\pi \frac{a+b}{3.6}\right) \tag{17}$$

We conducted three sets of simulations: low-speed, high-speed, and variable speed–amplitude combinations. The main findings are summarized below:

- **Low speeds (1–35 km/h)**: Vehicle body shows persistent oscillations stabilizing after 6s.
- **High speeds (50–250 km/h)**: The body settles faster due to higher frequency excitation, typically within 5s.
- **Amplitude sensitivity**: At fixed speeds, increasing road amplitude leads to significantly higher body and wheel displacements. Notably, at $h = 0.1$ m, wheel detachment from the ground becomes prominent.

These results suggest that both speed and road amplitude critically influence ride comfort and suspension load, with amplitude being the dominant factor in extreme responses.

For detailed plots and simulation configurations, see Appendix D.

3.3. Effect of a speed bump

To evaluate suspension performance under realistic urban conditions, we simulate the vehicle's response to two common UK speed bump designs: the **Watts hump** and the **Seminole hump**. These profiles were chosen for their regulatory relevance and differing geometric characteristics (parabolic vs. flat-topped).

Using a non-linear half-car model, we analyze body and wheel displacements over these profiles to assess how shock characteristics vary.

Key Observations

- **Watts profile**: Results in smooth, oscillatory responses as the car transitions over the bump. Front and rear axle impacts are separated in time, with a natural return to steady state.
- **Seminole profile**: Induces sharper initial displacements due to flat-top geometry, followed by damped oscillations as the car descends. The flat region yields reduced interaction duration compared to the Watts profile.

For detailed Simulink implementation parameters and full displacement plots, refer to Appendix C.6.

3.4. Sensitivity Analysis Parameters

A global sensitivity analysis was performed on all suspension parameters using two sampling methods: Monte Carlo and Sobol sequences. For each method, 100 samples were generated assuming a uniform distribution, as no prior information about parameter distributions was available[2].

Monte Carlo sampling provided a baseline for random exploration, while Sobol sampling—configured with a skip of 1, leap of 0, and Matoušek-Affine-Owen scrambling—offered improved space-filling properties and reproducibility via standard point ordering.

4. Results and Analysis

This section presents a systematic analysis of suspension parameter sensitivity and optimization, focusing on enhancing ride comfort. We employ both Monte Carlo and Sobol sampling methods to evaluate the influence of key parameters, using the step-response envelope as a consistent baseline.

[2]Uniform priors were selected due to lack of empirical distribution data.

Table 2. Sensitivity Analysis Using Uniformly Distributed, Independently Sampled Parameters (Assuming Rear Parameters $\leq$ Front Parameters

Parameters	**Bound**	**Justification**
$C_{s,Bump,f}$	[400,1500]	Soft to firm
$C_{s,Bump,r}$	[400,1500]	Soft to firm
$C_{s,Rebound,f}$	[800, 2500]	1.5–2× $C_{s,Bump,f}$
$C_{s,Rebound,r}$	[800, 2500]	1.5–2× $C_{s,Bump,r}$
I	[400,1500]	Mid-size car
$K_{s,r}$	[15k,50k]	Mid-size car
$K_{s,f}$	[15k,50k]	Mid-size car
$K_{s,stiff,r}$	[300k,900k]	10–30× $K_{s,r}$
$K_{s,stiff,f}$	[300k,900k]	10–30× $K_{s,l}$
K_t	[100k,300k]	Standard road tires
M_b	[500, 900]	Standard mass
M_w	[30,60]	Unsprung wheel mass
a	[0.8,1.5]	Logical wheel base
b	[0.8,1.5]	$a + b \approx \frac{1}{2}$ wheelbase

This allowed us to identify the most influential variables before applying BO. Final optimized values are listed in Appendix E.

Ride comfort was quantitatively assessed using the RMS of body acceleration. Given its frequency-rich characteristics, the sinusoidal road profile was selected as the optimization target environment, as established in the transient and bump tests in Section 3. These scenarios simulate some of the harshest realistic driving conditions, ensuring robustness in the resulting setup.

Parameter tuning was performed using MATLAB's **Response Optimizer**, with the **Step Envelope** tool employed to constrain bounce behavior within acceptable thresholds over time. This enabled precise tuning of damper characteristics (for both rebound and compression), as well as spring stiffness and nonlinear hardening behavior.

It should be noted that gradient-free optimization techniques, including BO, may converge to local rather than global optima due to limited sampling and model uncertainty.

4.1. Sensitivity Analysis

Monte Carlo-Based Sensitivity Analysis Figure 7 presents the results of sensitivity analysis using Monte Carlo sampling. The parameters' influence was quantified using correlation, standardized regression, and partial correlation methods.

The analysis highlights body mass M_b and wheel mass M_w as the most critical parameters influencing the vehicle's RMS body acceleration. A negative influence of Mb implies that increasing body mass reduces acceleration, enhanc-

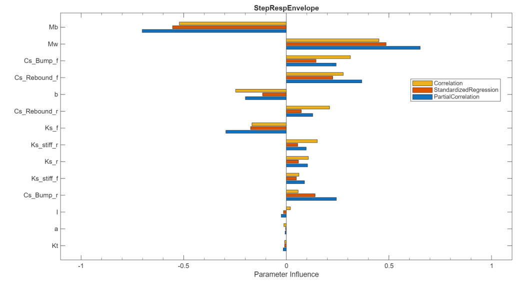


Figure 7. Monte Carlo sensitivity analysis of RMS_Acc_Body.

ing comfort. Conversely, an increase in wheel mass M_w correlates positively with increased acceleration, adversely affecting comfort. Secondary influences include front rebound damping, Rebound, $C_{s,Rebound,f}$ and front spring stiffness, $K_{s,f}$, indicating that suspension tuning at the front axle is critical.

Sobol-Based Sensitivity Analysis To further enhance the robustness of the sensitivity analysis, Sobol sampling was employed. Sobol sampling ensures better coverage and uniformity of parameter space, especially beneficial for high-dimensional analyses, avoiding the clustering seen with random sampling methods.

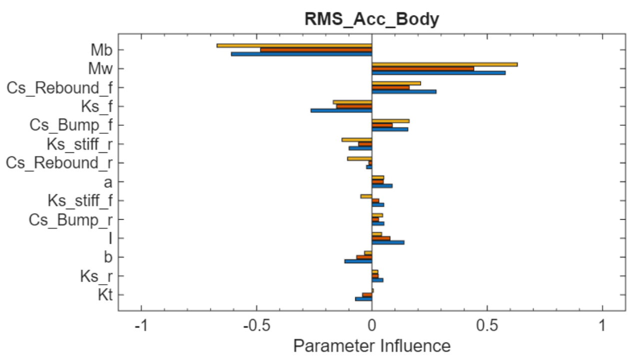


Figure 8. Sobol sensitivity indices for RMS_Acc_Body.

Consistent with Monte Carlo findings, Sobol indices (Figure 8) reveal body mass Mb and wheel mass Mw as dominant parameters. The increased clarity in the separation of parameter influences further underscores the advantage of using Sobol sampling. Additional parameters such as rear spring stiffness, $K_{s,r}$and rear rebound damping, Rebound, $C_{s,Rebound,r}$ show moderate importance, signifying the need for balanced tuning across both axles.

4.2. Comparison of Sampling Methods

Convergence Analysis A comparative analysis of the convergence behavior between Monte Carlo (Random) and Sobol methods is depicted in Figure 9. Convergence is measured by the best RMS acceleration achieved as a function of the number of samples evaluated.

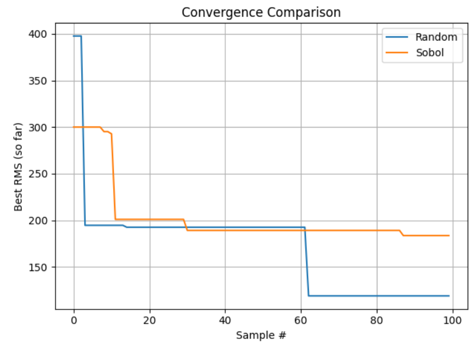


Figure 9. Convergence comparison of Monte Carlo and Sobol sampling methods.

Sobol sampling achieves more stable and rapid initial convergence, reaching an RMS value around 200 within approximately 20 samples. The Monte Carlo method exhibits slower initial convergence but achieves lower RMS values (below 150) after about 60 evaluations. This result suggests Sobol sampling offers efficient exploration initially, but Monte Carlo sampling can reach superior solutions given sufficient sampling time.

Feature Importance Analysis To further investigate differences between sampling methods, a Random Forest regression model was used to quantify feature importance (Figure 10).

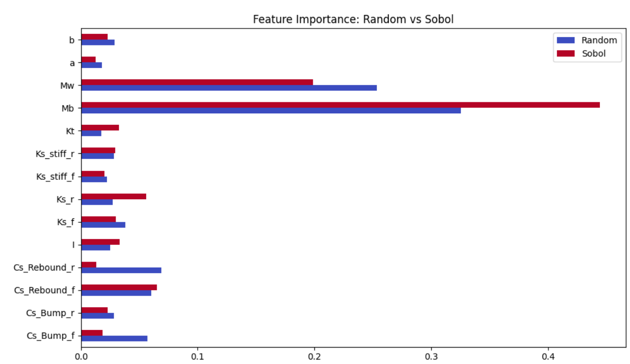


Figure 10. Feature importance derived from Random Forest regression using Monte Carlo (Random) vs. Sobol sampling.

Feature importance rankings are consistent across both methods. Body mass M_b and wheel mass M_w emerge clearly as the most influential parameters. Notably, the Sobol-based method reveals slightly stronger distinctions between primary and secondary influences, reflecting its more structured sampling.

4.3. Step Response Envelope Optimization

To optimize the vehicle suspension for improved ride comfort, we conducted an envelope optimization based on the vehicle's step response.

Figure 11 illustrates the step response of the suspension system before (black) (original parameters can be found in Appendix E.2 and after (blue) optimization. The optimized configuration demonstrates improved compliance with the envelope bounds and a clear reduction in body RMS acceleration. The optimization process converged within 20 minutes.

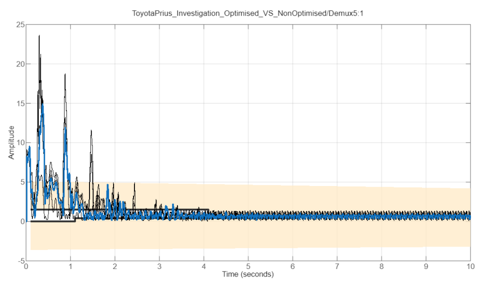


Figure 11. Step response comparison between non-optimized (black) and optimized (blue) suspension parameters: body displacement (top), pitch angle (middle), and body RMS acceleration (bottom).

The optimized parameters notably improved the dynamic response of the vehicle, as evidenced by:

- Reduced peak body displacement, demonstrating better shock absorption capability.
- A significant reduction in pitch angle oscillations, improving the vehicle's stability and passenger comfort.
- Considerably lower body RMS acceleration, indicating enhanced comfort and smoother ride quality.

Frequency Response Improvement The envelope optimization also achieved a notable reduction in the natural frequency of the vehicle's body response. Figure 12 shows that the optimized parameters successfully decreased the natural frequency from approximately 20 Hz (original parameters) to 8.1 Hz (optimized parameters), aligning better with typical human comfort frequencies.

4.4. Bayesian Optimization

Full Bayesian Optimization The BO was initially performed with all suspension parameters considered. Results are summarized in Table 8. The optimization significantly improved the suspension dynamics, lowering the frequency of body oscillations from approximately 20 Hz (original non-optimized system) to a notably lower value of 2.9 Hz.

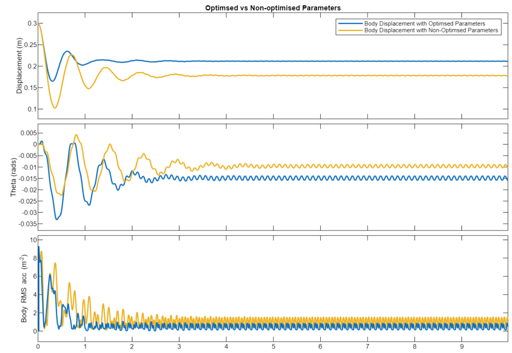


Figure 12. Reduction in RMS acceleration frequency from original 20 Hz (non-optimized) to 8.1 Hz (optimized).

The sweep was done in less than 10 minutes which shows a computational improvement to our previous method. The new parameters can be seen in Appendix E.3 and the minimum signal achieved by the objective function is 0.199.

It is essential to critically assess whether simulation results align with real-world mechanical principles and vehicle design conventions. A simulation, no matter how complex or well-constructed, must be validated against physically plausible behaviors to ensure credibility and usefulness in engineering applications. Parameters such as mass distribution, wheelbase length, and suspension dynamics should reflect established automotive design norms. This not only provides confidence in the model's predictive capability but also ensures that any subsequent optimization is grounded in physical reality rather than numerical artifacts.

Interpretation in Terms of Vehicle Geometry: The total wheelbase of the vehicle is $a + b = 2.1392\,\mathrm{m}$, indicating that the center of gravity (CoG) is located closer to the front axle:

$$\frac{a}{a+b} = \frac{1.1719}{2.1392} \approx 54.8\%, \qquad \frac{b}{a+b} \approx 45.2\%$$

This forward-biased distribution is typical of front-engine vehicles such as the Toyota Prius, where the majority of the mass—engine, transmission, and drivetrain—is situated toward the front. Such a configuration supports the realism of the simulation results and confirms that the model adheres to expected vehicle dynamics.

Figure 13 illustrates the convergence behavior, demonstrating a rapid initial reduction in the objective value followed by gradual improvements until convergence.

Figure 14 compares the optimized (blue) and non-optimized (yellow) responses in displacement, pitch angle, and body RMS acceleration, highlighting substantial reductions in amplitude and frequency of oscillations.

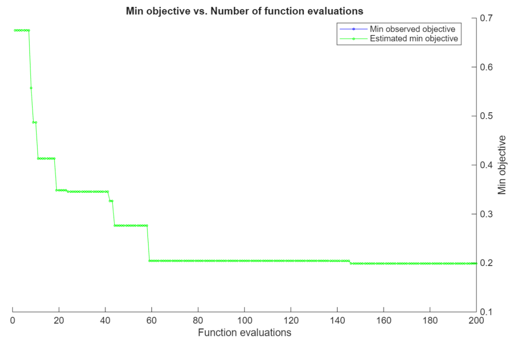


Figure 13. Objective function convergence during full Bayesian Optimization.

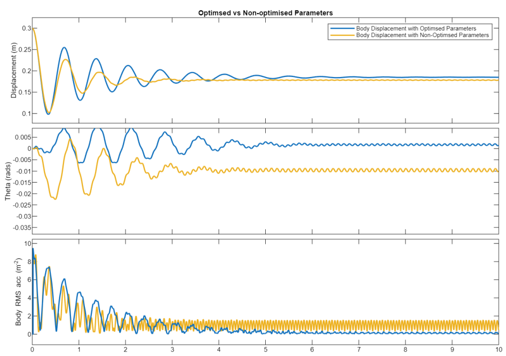


Figure 14. Response comparison for optimized versus non-optimized suspension parameters.

BO with Selected Influential Parameters Following insights from sensitivity analyses, we performed a targeted BO, excluding the less influential parameters (moment of inertia I, and distances a, b). Values for I, a, b and were fixed according to the previous optimized results. This refined optimization strategy resulted in further reducing the system's oscillation frequency down to 2.7 Hz.

Figure 15 clearly demonstrates effective convergence, emphasizing the benefit of reducing the optimization problem's dimensionality to avoid local optima.

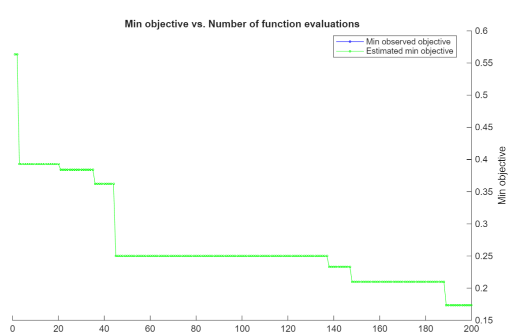


Figure 15. Convergence of the Bayesian Optimization without parameters . Minimum value obtained at 0.17

Figure 16 confirms further improvement in dynamic response. The optimized suspension exhibits significantly reduced oscillation amplitudes and quicker damping of transient behavior, suggesting enhanced passenger comfort and stability.

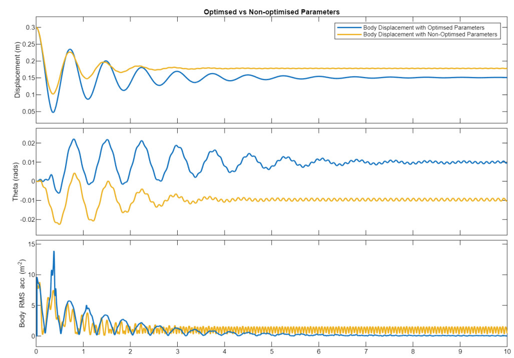


Figure 16. Optimized vs. non-optimized response excluding parameters I, a, b.

5. Conclusion

This study demonstrated the effectiveness of sensitivity analysis and Bayesian optimization in tuning a half-car suspension model for improved ride comfort. By targeting critical parameters and evaluating dynamic performance under varying road conditions, we achieved significant reductions in body acceleration and improved damping response within human comfort frequency ranges from 20 to 2.7Hz.

6. Limitations and Future Work

While the model captures vertical dynamics and core suspension behavior, it omits tyre handling characteristics, lateral dynamics, and detailed noise–vibration–harshness sources. Future work should integrate tyre–road interaction models, auditory vibration sensitivity, and aerodynamic or powertrain-induced noise to provide a more holistic optimization framework for passenger comfort.

Acknowledgements

Thank you to Dr. Carl Hendrick and Prof. Neil Lawrence

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A. Nomenclature

Table 3. List of symbols

Symbol	Description	Unit
a	Distance from f to CoG	m
b	Distance from CoG to r	m
CoG	Center of Gravity	-
$C_{s\_contraction}$	Damping Rate in bumps	Ns/m
$C_{s\_extension}$	Damping Rate in rebound	Ns/m
C_{s_nc}	Damping Rate Normal Conditions	Ns/m
f	Front Axle	-
f_r	Frequency Rate	Hz
F_f	Front Axle Force	N
F_r	Rear Axle Force	N
FBD	Free Body Diagram	-
g	Gravity	m/s^2
I	Moment of Inertia	kg/m^2
K_s	Spring Stiffness	N/m
K_{s_stiff}	Spring Stiffness Hardening Effect	N/m
K_t	Tyre Stiffness	N/m
M_b	Body Mass	kg
M_w	Wheel Mass	kg
ω	Angular Velocity	Rad/s
r	Rear Axle Force	-
S_b	Body Displacement	m
$\dot{S}_b$	Body Velocity	m/s
$\ddot{S}_b$	Body Acceleration	m/s^2
S_w	Wheel Displacement	m
$\dot{S}_w$	Wheel Velocity	m/s
$\ddot{S}_w$	Wheel Acceleration	m/s^2
t	Time	s
Θ	Pitch Angle	$degrees$

B. Test Benches

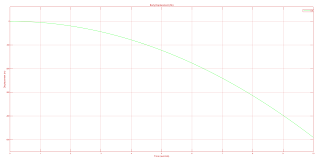
Body Subsystem In order to achieve success in our ***Fall Under Gravity***, it is imperative that the resolved vertical forces we have derived align with the accurate calculation of body displacement, velocity, and acceleration. These aspects can be defined by the following equations:

$$\dot{S}_B = -g \tag{18}$$

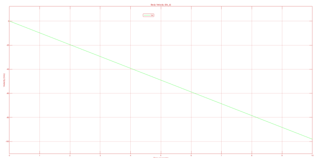
$$\dot{S}_B = \int \ddot{S}_B \, dt = -gt \tag{19}$$

$$S_B = \int \dot{S}_B \, dt = -\frac{gt^2}{2} \tag{20}$$

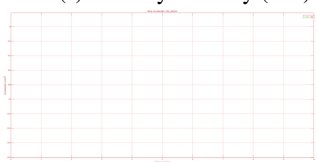
It can be observed that Eq 18, Eq 19, and Eq 20 correspondingly align with the output displayed in Fig 17a, Fig 17b, and Fig 17c:



(a) Fall-Under-Gravity Test - Time (s) vs Body Displacement (m)



(b) Fall-Under-Gravity Test - Time (s) vs Body Velocity (m/s)



(c) Fall-Under-Gravity Test - Time (s) vs Body Acceleration m/s^2 (it will just display a green constant dot at g=9.81

Figure 17. Fall-Under-Gravity Test for Body

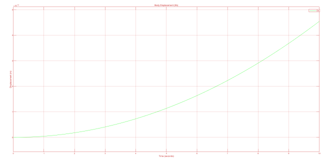
In order to achieve success in our ***Balance Weight Test***, we can resolve it analytically by computing the body mass distribution, as depicted in Eq 21 and Eq 22, consequently yielding the vertical forces:

$$M_{b,f} = M_b \frac{b}{a+b} \tag{21}$$

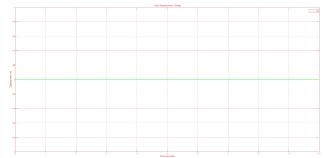
$$M_{b,r} = M_b \frac{a}{a+b} \tag{22}$$

$$\tag{23}$$

It can be observed in Fig 18a and Fig 18b that there is a negligible variation in body displacement (10^{-14}), which effectively aligns with the equations mentioned earlier for body displacement and the constant pitch angle over time:



(a) Balance Weight Test - Time (s) vs Body Displacement (m)



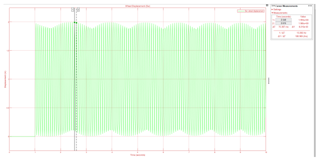
(b) Balance Weight Test - Time (s) vs Theta Displacement (m)

Figure 18. Balance Weight Test for Body

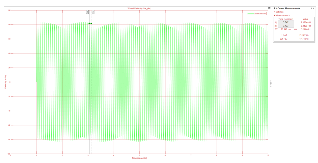
Wheels Subsystems The tests for both **Fall Under Gravity** and **Balance Weight Test** are anticipated to yield the same results as the body test described above since they are subjected to identical vertical forces. Therefore, kindly refer to the aforementioned graphs to comprehend the behavior of the wheel under these test conditions, which remain consistent.

The accuracy of our ***Step Input Test*** was successfully assessed by manually calculating the frequency and comparing it with the frequency of our wheel displacement using the *Cursor Measurement* tool. The mean displacement and velocity were determined to be 1 and 0, respectively. As shown in Eq 24, our computed frequencies of 13.265 Hz and 13.167 Hz (observed in Fig 19a and Fig 19b) closely approximate the numerical frequency of 13.315 Hz:

$$f = \frac{\omega}{2\pi} = \frac{\sqrt{\frac{K_t}{M_w}}}{2\pi} = \frac{\sqrt{\frac{14*10^4}{20}}}{2\pi} = 13.315Hz \tag{24}$$



(a) Displacement — frequency 13.265 Hz



(b) Velocity — frequency 13.167 Hz

Figure 19. Wheel Subsystem Step Input Response: Displacement and Velocity

Suspension Subsystem We have generated 16 combinations of binary values within the array shown in Eq 25 for the four variables described in Eq 8. Subsequently, we computed the output of these combinations to facilitate a comparison with the analytical solution. It is important to note that we exclusively utilized a linear model with constant parameters, where K_s was set at 20,000 N/m and Cs_{nc} at 1000 Ns/m, to perform this task.

$$\left[\dot{S}_{b(f/r)},\ \dot{S}_{b(r)},\ \dot{S}_{w(f/r)},\ \dot{S}_{w(r)}\right] \tag{25}$$

The following binary inputs combination and outputs were computed in Table 4:

Binary Inputs				**Calc. Force (N)**	**Analytical (N)**
$\dot{S}_b$	S_b	$\dot{S}_w$	S_w		
0	0	0	0	0	0
0	0	0	1	20000	20000
0	0	1	0	1000	1000
0	0	1	1	21000	21000
0	1	0	0	-20000	-20000
0	1	0	1	0	0
0	1	1	0	-19000	-19000
0	1	1	1	1000	1000
1	0	0	0	-1000	-1000
1	0	0	1	19000	19000
1	0	1	0	0	0
1	0	1	1	20000	20000
1	1	0	0	-21000	-21000
1	1	0	1	-1000	-1000
1	1	1	0	-20000	-20000
1	1	1	1	0	0

Table 4. Suspension model I/O test results comparing computed and analytical forces.

Half Car Subsystem

$$S_w = -\frac{(M_b + 2M_w)g}{2K_t} \tag{26}$$

$$S_b = \left(-\frac{M_b g}{2K_s}\right) + S_w \tag{27}$$

$$w = \sqrt{\frac{2K_t}{(2M_w + M_b)}} \tag{28}$$

$$w = \sqrt{\frac{2K_s}{M_b}} \tag{29}$$

$$\tag{30}$$

In the context of the **Balance Weight Test**, our objective is to verify that both the body and the wheels experience free fall under gravity and subsequently oscillate until they attain a state of equilibrium and rest. This equilibrium state is accomplished through the balancing of suspension forces and tire stiffness between the body and the wheel subsystems.

We have undertaken the analytical calculation of the settling displacement values for both the body and the wheels (both front and rear). These calculations are based on the equations detailed in Table **??**, along with Eq 21 and Eq 22. The obtained analytical results are as follows:

$$S_w = \frac{-[M_b + 2M_w]g}{2K_t} = -0.0259 \text{ m} \tag{31}$$

$$S_b = \frac{-[M_b g]}{2K_s} + S_w = -0.1976 \text{ m} \tag{32}$$

$$\tag{33}$$

For the front and rear:

$$S_{w,r} = \frac{-[M_b\left(\frac{a}{a+b}\right) + M_w]g}{K_t} = -0.0246 \text{ m} \tag{34}$$

$$S_{w,f} = \frac{-[M_b\left(\frac{b}{a+b}\right) + M_w]g}{K_t} = -0.0272 \text{ m} \tag{35}$$

$$S_{b,r} = \frac{-[M_b\left(\frac{a}{a+b}\right)]g}{K_s} + S_{w,r} = -0.1873 \text{ m} \tag{36}$$

$$S_{b,f} = \frac{-[M_b\left(\frac{b}{a+b}\right)]g}{K_s} + S_{w,f} = -0.2079 \text{ m} \tag{37}$$

$$\tag{38}$$

In Fig 20, we can discern the oscillatory behavior exhibited by both the wheel and the body, which is primarily attributed to the damping effect. It is noteworthy that the steady-state displacement aligns with the previously mentioned analytical values.

Moreover, we can observe that the distribution of vehicle mass has led to greater settling displacements at the front axle. This phenomenon is a result of the front spring needing to exert a higher force to establish equilibrium.

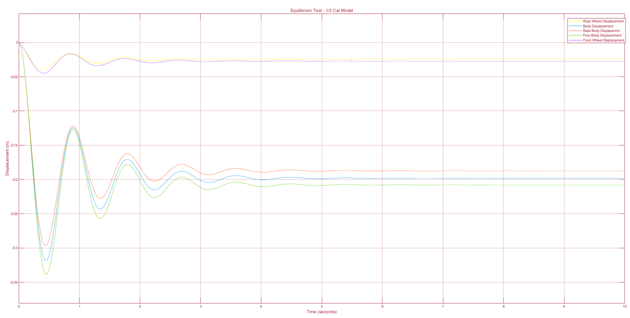
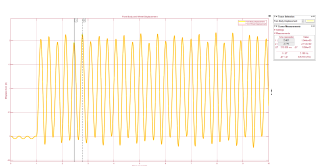


Figure 20. Half Car Model Balance Test

In the context of the ***Wheel Bounce Test***, our aim is to simulate a rigid rod suspension subjected to a simultaneous step road input (0-1). As previously described, this can be accomplished by selecting substantial values for stiffness (K_s, e.g., 10^9) and damping (C_s, e.g., 2×10^{10}). These parameter choices effectively minimize suspension displacement, resulting in a nearly rigid body configuration.

The results for both the front and rear wheel and body displacements can be observed in Fig 21a and Fig 21b, respectively:



(a) Front wheel and body displacement

(b) Rear wheel and body displacement

Figure 21. Half-Car Wheel Bounce Test showing front and rear wheel/body displacement under a step input with high stiffness and damping.

From the figures above, it is evident that we have successfully implemented a single rigid rod suspension, resulting in identical outputs for both body and wheel displacements, whether at the rear or front. The graph exhibits a wheel detachment, which was incorporated for testing purposes and to enhance our results, showcasing the oscillatory bounce and rebound motion.

Furthermore, we can observe that the analytical frequency, as depicted in Eq 40, closely aligns with the frequencies illustrated in the graphs above (front = 3.166 Hz and rear = 3.196 Hz):

$$\omega = \sqrt{\frac{2K_t}{2M_w + M_b}} = \sqrt{\frac{2 \times 14 \times 10^4}{2 \times 20 + 700}} = 19.452 rad/s \tag{39}$$

$$f = \frac{\omega}{2\pi} = \frac{19.452}{2\pi} = 3.0959 Hz \tag{40}$$

In the context of the ***Spring Bounce Test***, we have implemented a step road input (0-1) while employing substantial tire stiffness (K_t, e.g., 14×10^7) and a low wheel weight (M_w, e.g., 0.0001 kg). This setup results in the oscillatory behavior depicted in Fig 22 for the Zero Damping scenario:

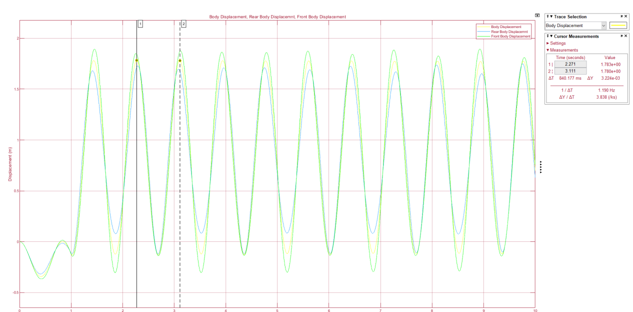


Figure 22. Half Car - Spring Bounce Test - Body Displacement for high K_t and low M_w

We can relate the accuracy of our computational method's frequency (e.g., 1.190 Hz) by analytically calculating the frequency as indicated in Eq 42:

$$\omega = \sqrt{\frac{2K_s}{M_b}} = \sqrt{\frac{2 \times 2 \times 10^4}{700}} = 7.56 rad/s \tag{41}$$

$$f = \frac{\omega}{2\pi} = \frac{7.56}{2\pi} = 1.2031 Hz \tag{42}$$

We have also conducted a scenario (Fig 23) involving Non-Zero Damping, with parameters such as Cs_{nc} set to 1000

Ns/m. This scenario illustrates how the damping effect causes oscillations in the body to gradually decay over time.

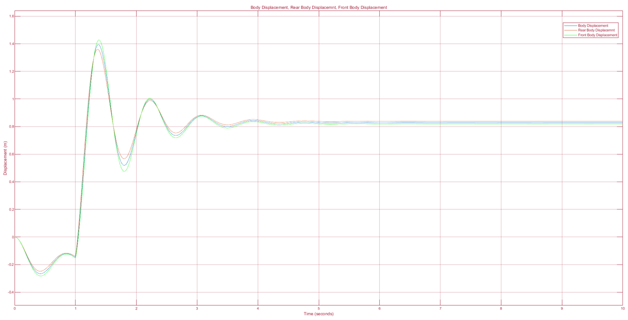
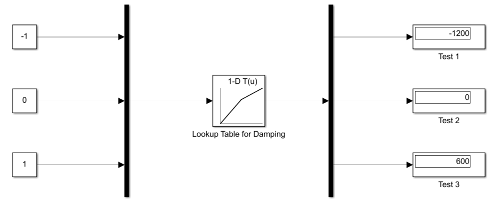
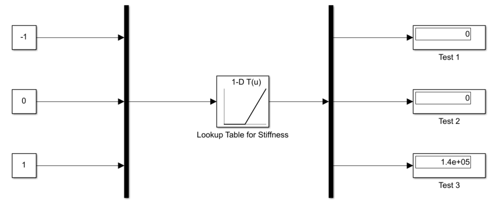


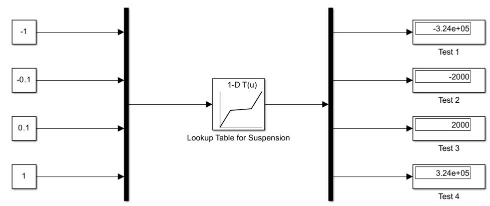
Figure 23. Half Car - Spring Bounce Test - Body Displacement for high K_t and low M_w with Non-Zero Damping



(a) Damping Lookup Table



(b) Stiffness Lookup Table



(c) Suspension Force Output

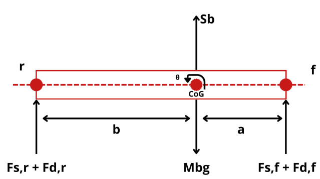
Figure 24. Lookup Table verification for damping, stiffness, and force computation.

Non-Linear Lookup Table Testing To incorporate non-linear properties into the model, we used Lookup Tables to define:

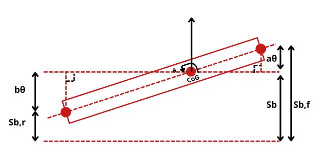
- Damping asymmetry between compression and extension
- Progressive (hardening) spring stiffness

We designed a dedicated test bench in Simulink to ensure the expected performance of each lookup table across a range of inputs.

C. Half-Car Model - Simulink



(a) Free Body Diagram Pitch And Roll at rest



(b) Free Body Diagram Pitch And Roll at a small angle change

Figure 25. Free Body Diagram of a Car's Body in two states: (a) at rest, and (b) experiencing a small angular change. In (a), the car is shown in a static equilibrium with forces $F_{s,f} + F_{d,r}$ and $F_{s,f} + F_{d,f}$ acting at the front and rear respectively, balanced by the gravitational force M_{bg} acting through the center of gravity. The distances from the center of gravity to the front and rear axles are denoted by a and b. In (b), the car is depicted with an angular displacement θ, causing a shift in the suspension forces which now act at $S_{b,f}$ and $S_{b,r}$, illustrating the body roll dynamics. This diagram is crucial for analyzing the effects of forces and moments on the car's pitch and roll during maneuvers.

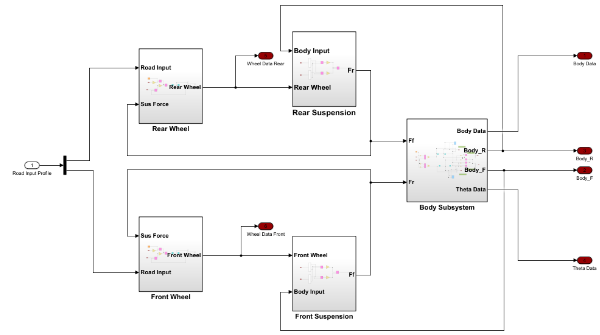


Figure 26. Simulink Model of Half Car Dynamics. The diagram depicts a system-level overview of a half car model used for dynamic analysis. Road input profiles are fed into the front and rear wheel subsystems, influencing the suspension forces (Sus Force) acting on each wheel. The 'Rear Wheel' and 'Front Wheel' blocks receive road inputs and suspension forces to compute the wheel dynamics. These are then directed to the 'Rear Suspension' and 'Front Suspension' blocks, respectively, which provide the body inputs for the 'Body Subsystem'. The 'Body Subsystem' integrates the forces from the front and rear to produce body data (displacement, velocity) and rotational data (theta). Outputs are depicted for body displacement and theta, reflecting the car's response to road irregularities, essential for analyzing ride quality and handling performance.

C.1. Body Subsystem

The physical implementation of the body module (Fig 27) has been realized in MATLAB Simulink and is available for reference in Appendix:

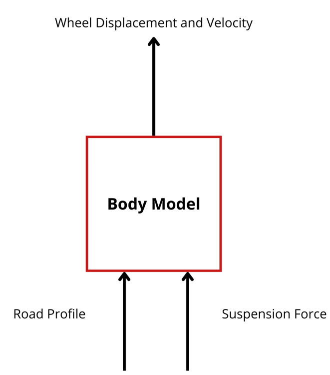


Figure 27. FBD Vehicle Body

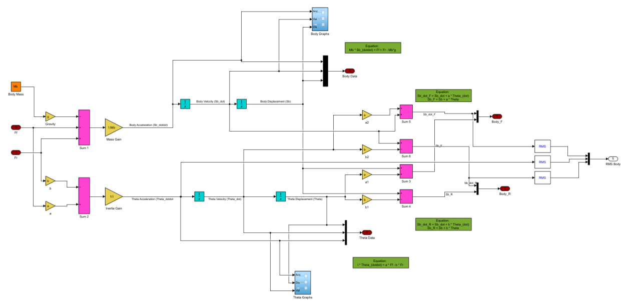


Figure 28. Body Subsystem

C.2. Wheel Subsystem

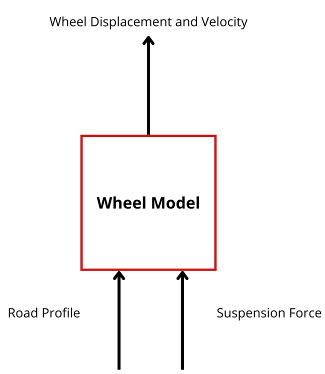


Figure 29. FBD Vehicle Wheel

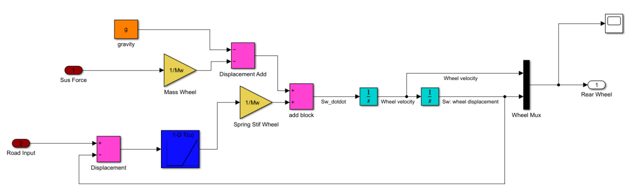


Figure 30. Wheel Subsystem

C.3. Suspension Subsystem

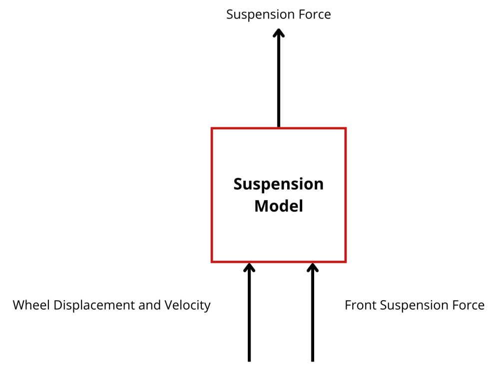


Figure 31. FBD Vehicle Suspension

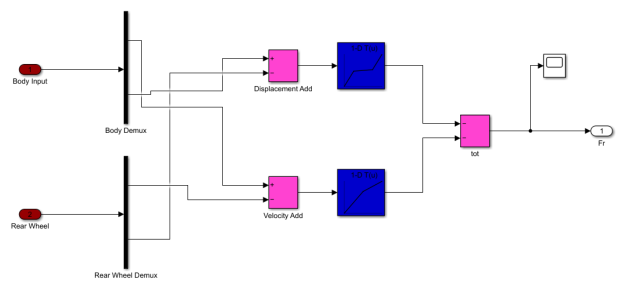


Figure 32. Suspension Subsystem

C.4. Non-Linear vs Linear Comparison

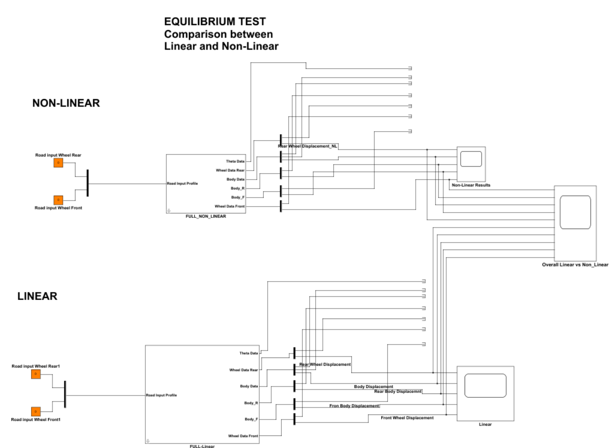


Figure 33. Non-Linear vs Linear Comparison

C.5. Non-Linear Wheel Detachment

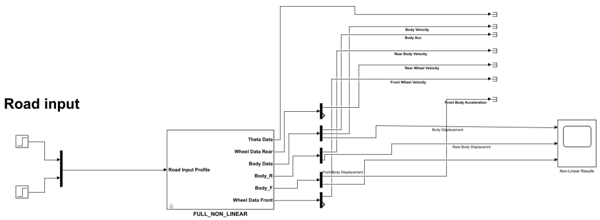


Figure 34. Non-Linear Wheel Detachment

C.6. Speed Bump Profiles and Simulation Results

Simulink Parameters and Profile Design The Watts and Seminole profiles were created using step and sine wave blocks. Their geometry and simulation conditions are summarized below.

Table 5. Speed Bump Simulation Parameters

Parameter	Symbol	Value
Shift	n	10
Bump Length	L	6.7m
Speed	v	8m/s
Front-to-CoG distance	a	1.35m
Rear-to-CoG distance	b	1.35m
Bump Height	h	0.3m

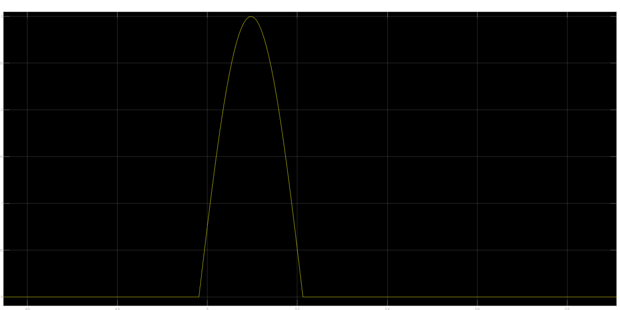


Figure 35. Watts Hump Geometry

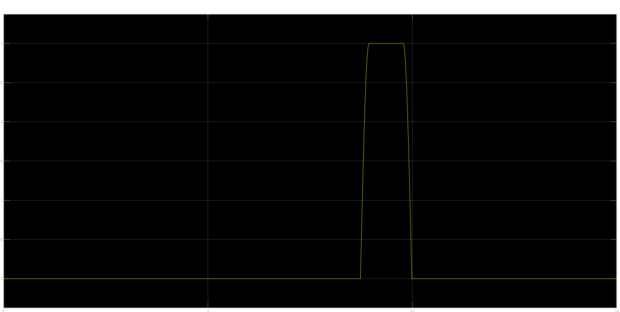


Figure 36. Seminole Hump Geometry

Profile Geometry

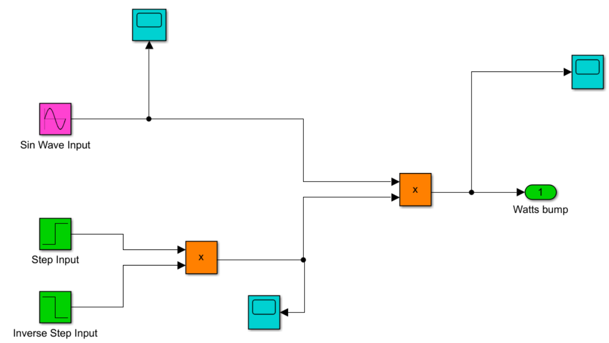


Figure 37. Watts Hump Simulink Implementation

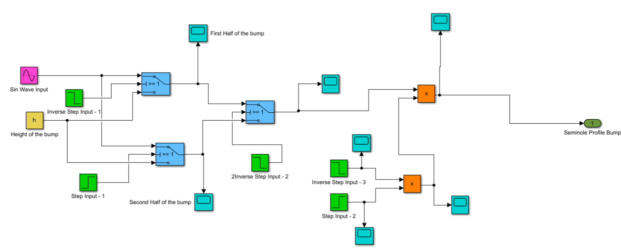


Figure 38. Seminole Hump Simulink Implementation

Simulink Implementations

Simulation Results: Displacements **Watts Profile Results:**

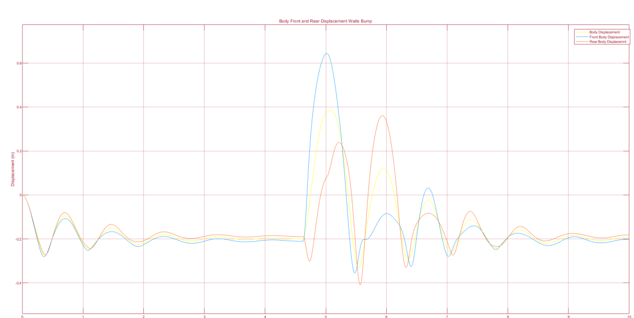


Figure 39. Body Displacement – Watts Profile

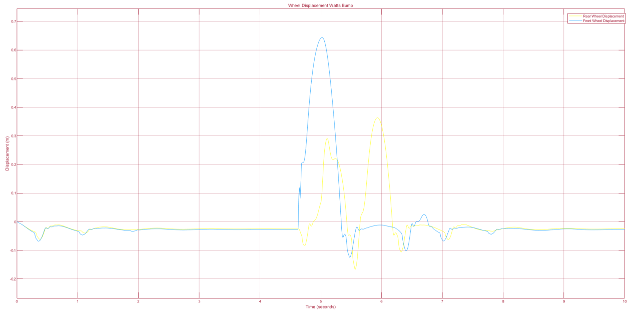


Figure 40. Wheel Displacement – Watts Profile

Seminole Profile Results:

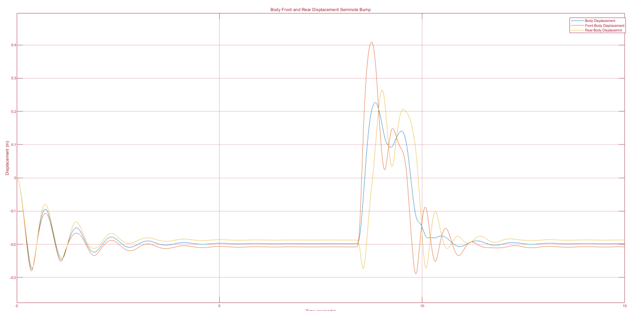


Figure 41. Body Displacement – Seminole Profile

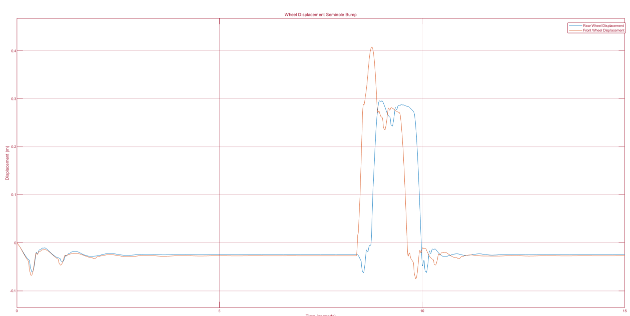


Figure 42. Wheel Displacement – Seminole Profile

D. Transient Response Simulations

D.1. Road Input Definition

The sinusoidal road profile used in Simulink is defined by:

$$r = h \sin\left(2\pi \frac{v}{3.6} t + 2\pi \frac{a+b}{3.6}\right) \tag{43}$$

D.2. Test Configuration

The following test scenarios were simulated:

Table 6. Transient Response Test Bench

Test	**Speed (km/h)**	**Amp (m)**	**Figs**
Low Speed	1–35	0.01	Fig 43, Fig 44
High Speed	50–250	0.01	Fig 45, Fig 46
Variable Speed/Amplitude	1–250	0.01–0.1	Fig 47, Fig 48

D.3. Results and Plots

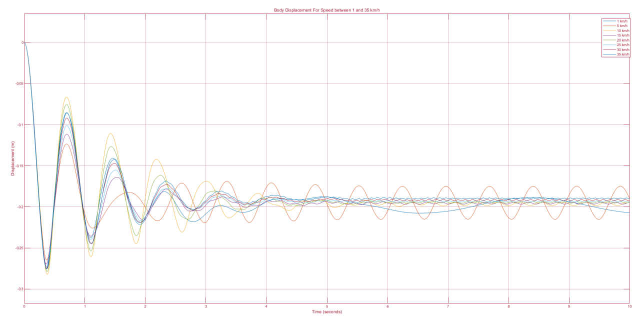


Figure 43. Body Displacement – Low Speed Test

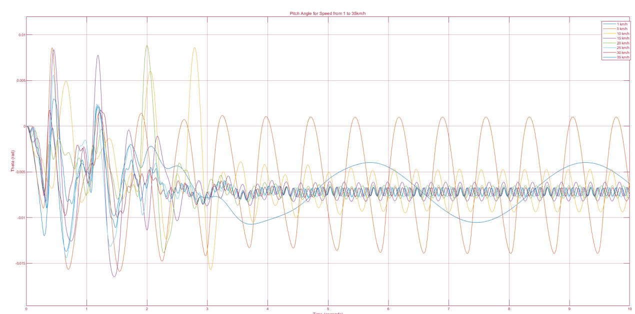


Figure 44. Pitch Angle – Low Speed Test

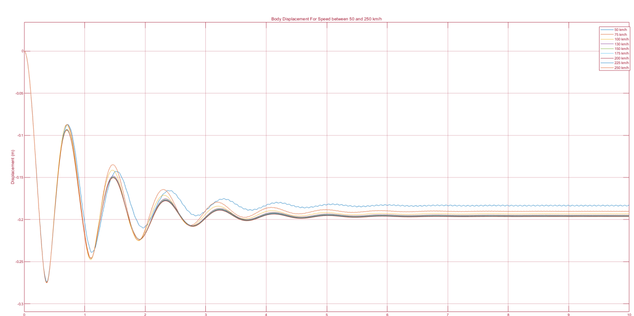


Figure 45. Body Displacement – High Speed Test

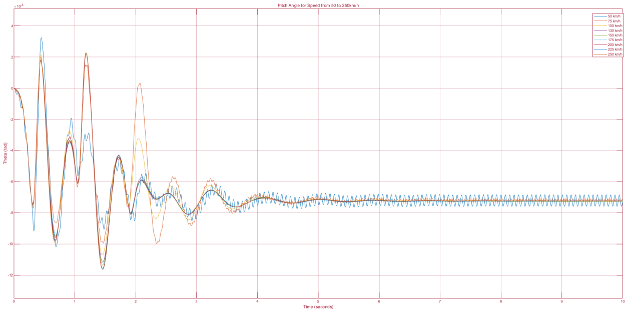


Figure 46. Pitch Angle – High Speed Test

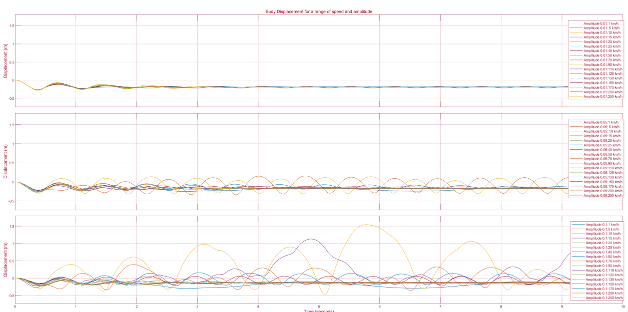


Figure 47. Body Displacement – Variable Speed and Amplitude

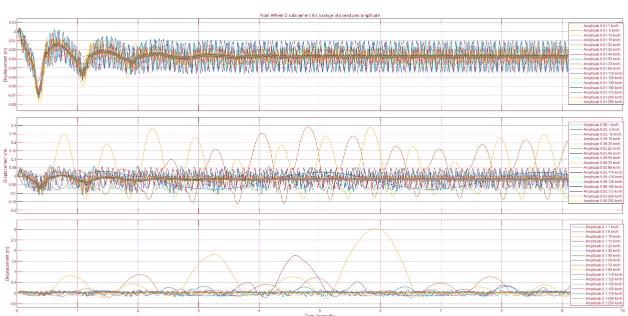


Figure 48. Wheel Displacement – Variable Speed and Amplitude

E. Results

E.1. Original Constructor Values for the Toyota Yaris

This is the original constructor values used as a baseline for our experiments:

Table 7. Initial Estimated Constructor Values for the Toyota Yaris

Parameters	Values
$C_{s,Bump,f}$	800
$C_{s,Bump,r}$	700
$C_{s,Rebound,f}$	1400
$C_{s,Rebound,r}$	1300
I	800
$K_{s,r}$	28000
$K_{s,f}$	31000
$K_{s,stiff,r}$	$20 \times K_{s,r}$
$K_{s,stiff,f}$	$20 \times K_{s,f}$
K_t	200000
M_b	650
M_w	35
a	1
b	1.3

E.2. Step Response Envelope - Parameters

Parameter	Best Observed Value	Units
Ks_f	29597	N/m
Ks_r	30199	N/m
$Cs_{Rebound\_f}$	1482	Ns/m
Cs_{Bump_f}	819	Ns/m
$Cs_{Rebound\_r}$	1315	Ns/m
Cs_{Bump_r}	698	Ns/m
Kt	211660	N/m
I	800	kg·m^2
Mb	470	kg
Mw	35	kg
a	0.98	m
b	1.47	m
Ks_{stiff_f}	620000	N/m
Ks_{stiff_r}	560000	N/m

E.3. Full BO - Parameters

Parameter	Best Observed Value	Units
Ks_f	47325	N/m
Ks_r	24841	N/m
$Cs_{Rebound\_f}$	1003.4	Ns/m
Cs_{Bump_f}	422.93	Ns/m
$Cs_{Rebound\_r}$	880.62	Ns/m
Cs_{Bump_r}	591.52	Ns/m
Kt	2.89×10^5	N/m
I	1420.9	kg·m^2
Mb	791.39	kg
Mw	34.89	kg
a	1.1719	m
b	0.9673	m
Ks_{stiff_f}	4.33×10^5	N/m
Ks_{stiff_r}	6.83×10^5	N/m

Table 8. Full BO

E.4. Restricted BO - Parameters

Parameter	Best Observed Value	Units
Ks_f	47134	N/m
Ks_r	24862	N/m
$Cs_{Rebound\_f}$	922.07	Ns/m
Cs_{Bump_f}	876.76	Ns/m
$Cs_{Rebound\_r}$	810.52	Ns/m
Cs_{Bump_r}	1098.8	Ns/m
Kt	205440	N/m
Mb	895.95	kg
Mw	39.33	kg
Ks_{stiff_f}	824480	N/m
Ks_{stiff_r}	309160	N/m

Table 9. Full BO without parameters